

Assignment - 1

* operating system

OS is system software that manages computer hardware & software resources.

* Types of operating system

1. Batch of operating system

A. batch operating system
executes similar jobs in batches without user interaction

Use : -
- payroll processing
- Bank transaction
- Billing system

Advantages : -

- Efficient for large repetitive tasks
- Reduces idle time
- Easy job management

Disadvantages :-

- No direct user interaction
- Long waiting time
- Difficult to debug

2. Time sharing OS

Use :-

- Multiuser system.
- Universities
- office networks

Advantages :-

- fast response time.
- supports multiple users
- Better CPU utilization

Disadvantages :-

- security issues
- High maintenance
- system failure affects all users

Distributed operating system

Uses :-

- cloud computing
- scientific research
- large organization

Advantages :-

- Resource sharing
- High performance
- Reliable

Disadvantages :-

- complex design
- Expensive setup
- Network dependency

Network operating system

USES :-

- File sharing
- printer sharing
- ~~Network dependency~~
- office networks

Advantages :-

- Centralized management
- Easy resource sharing
- Better security

Disadvantages :-

- If server failure affects the network
- Costly
- Requires skilled administrator

Real-time operating system :-

Uses :-

- Medical equipment
- Air traffic control
- Robotics

Advantages :-

- Fast response
- High reliable
- Accurate performance

Disadvantages :-

- Expensive
- Complex programming
- ~~Not~~ limited multitasking

Multiprocessing operating system

Uses :-

- Scientific computing
- Engineering application
- Data centers.

Advantages :-

- High speed
- Better performance
- Increased reliability

Disadvantages :-

- Expensive hardware
- Complex design
- Difficult maintenance

→ Multitasking operating system

Uses :-

- Personal computers
- Laptops
- Smartphones

Advantages :-

- Saves time
- Better productivity
- Efficient CPU usage

Disadvantages :-

- Requires more memory
- may slow down with many programs
- Complex management

System calls :- System call is an interface betⁿ a user program & the OS

Types of system calls

1. Process control system calls :-
These system calls are used to create, execute, terminate & manage processes.
2. File management system calls :-
These system calls are ~~used~~ to create, open, read, write, close, delete, & manage files.

3 Device Management system calls:-

These system calls are used to control and communicate with hardware devices.

4. Information Maintenance system calls:-

These system calls are used to get or set system and process information.

5. Communicate system calls:-

These system calls allow communication between different processes.

6. protection system calls:-

These system calls provide security by controlling access to system resources.

* Definitions of system calls.

① fork ()

→ Create a new child process by copying the parent process

2) `vfork()`

create a child process that shares the parent's memory until it creates a new program or exits.

3) `clone()`

create a child process that shares the parent's memory until it executes a new program or exits.

4) `execve()`

Replace the current process with a new program.

5) `exit()`

Terminates a process after performing cleanup.

6) `_exit()`

immediately terminates a process without cleanup

7) `wait()`

waits until any child process finishes execution

8) wait pid ()
waits for a specific child process to finish

9) waitid ()
waits for a specific child finish

10) kill ()
sends a signal to a process

11) getpid ()
Returns the process id (pid) of the current process

12) getppid ()
Returns the parent process id (ppid)

13) nice ()
changes the scheduling priority of a process.

14) setpriority ()
set the priority of process.

- 16) `open()`
opens an existing file.
- 17) `creat()`
create a new file.
- 18) `close()`
close a new file.
- 19) `read()`
Read data from a file.
- 20) `write()`
writes data to a file.
- 21) `pread()`
Reads data from a specific file.
- 22) `pwrite()`
writes data to a specific the position without changing pointer.
- 23) `lseek()`
changes the current position of a file pointer.
- 24) `stat()`
Returns information about a file.

25) `fstat()`

Returns information about a
an open file.

26) `lstat()`

Return information about a
symbolic link

27) `access()`

Checks the access permission of
a file.

28) `dup()`

Creates a duplicate of a file
descriptor

29) `dup2()`

Duplicates a file descriptor to
a specified descriptor.

30) `fcntl()`

performs operations on file
descriptors, such as changing
flags or locking.

31) `fsync()`

writes all modified file data &
meta-data to the disk

32) `fdatasync()`

write modified data to the
disk but minimizes metadata
updates.

33) `truncate()`
changes the size of

34) `ftruncate()`
- changes the size of a
descriptor.